Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.503200
<D> / site : 0.250533

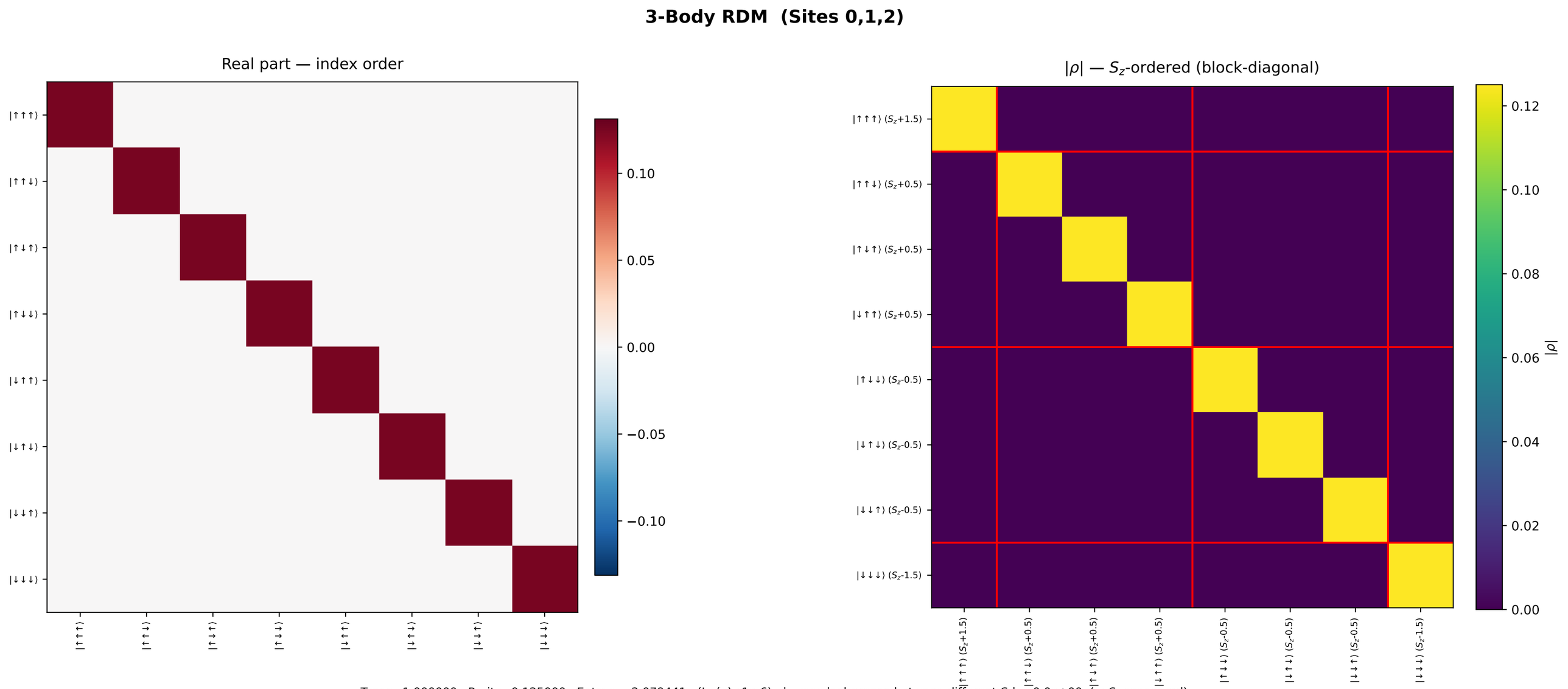
THIS PDF

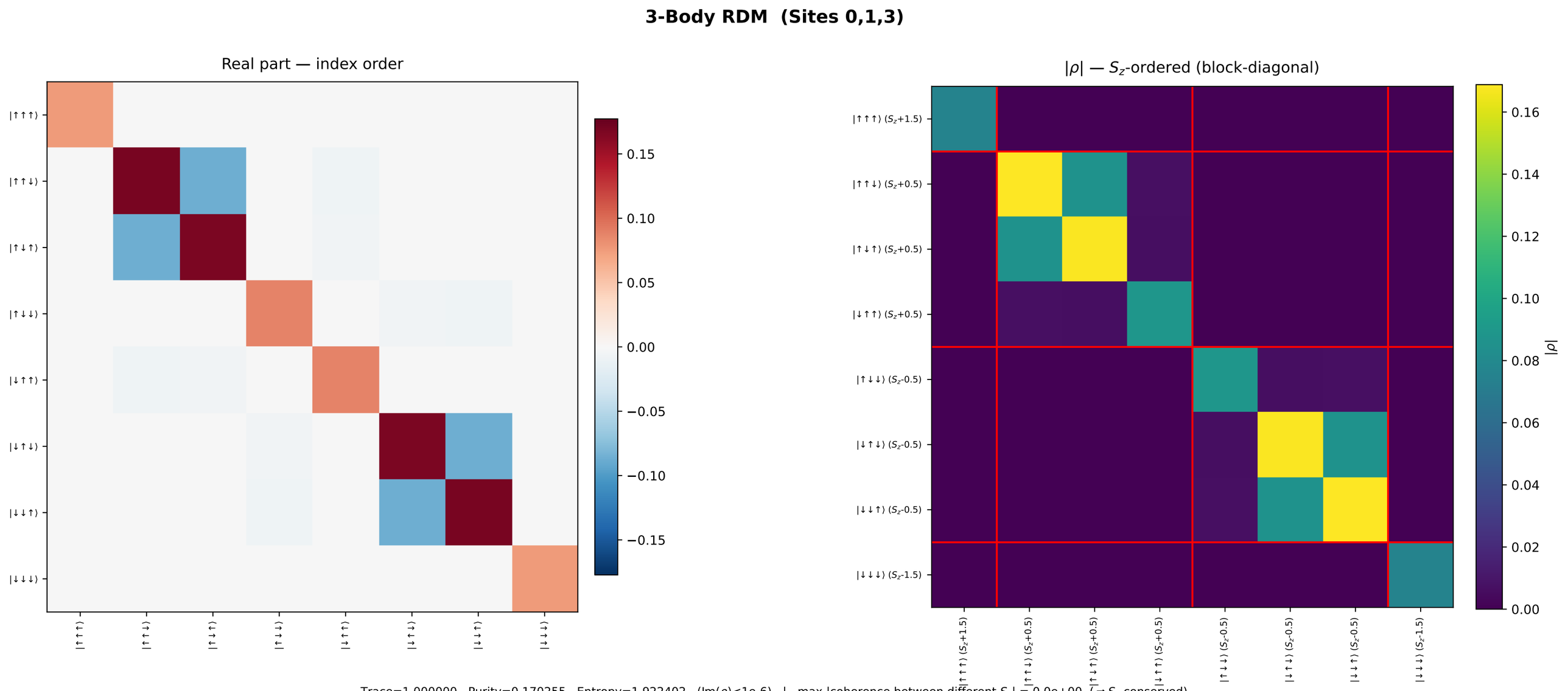
Body order : 3
Clusters   : 42

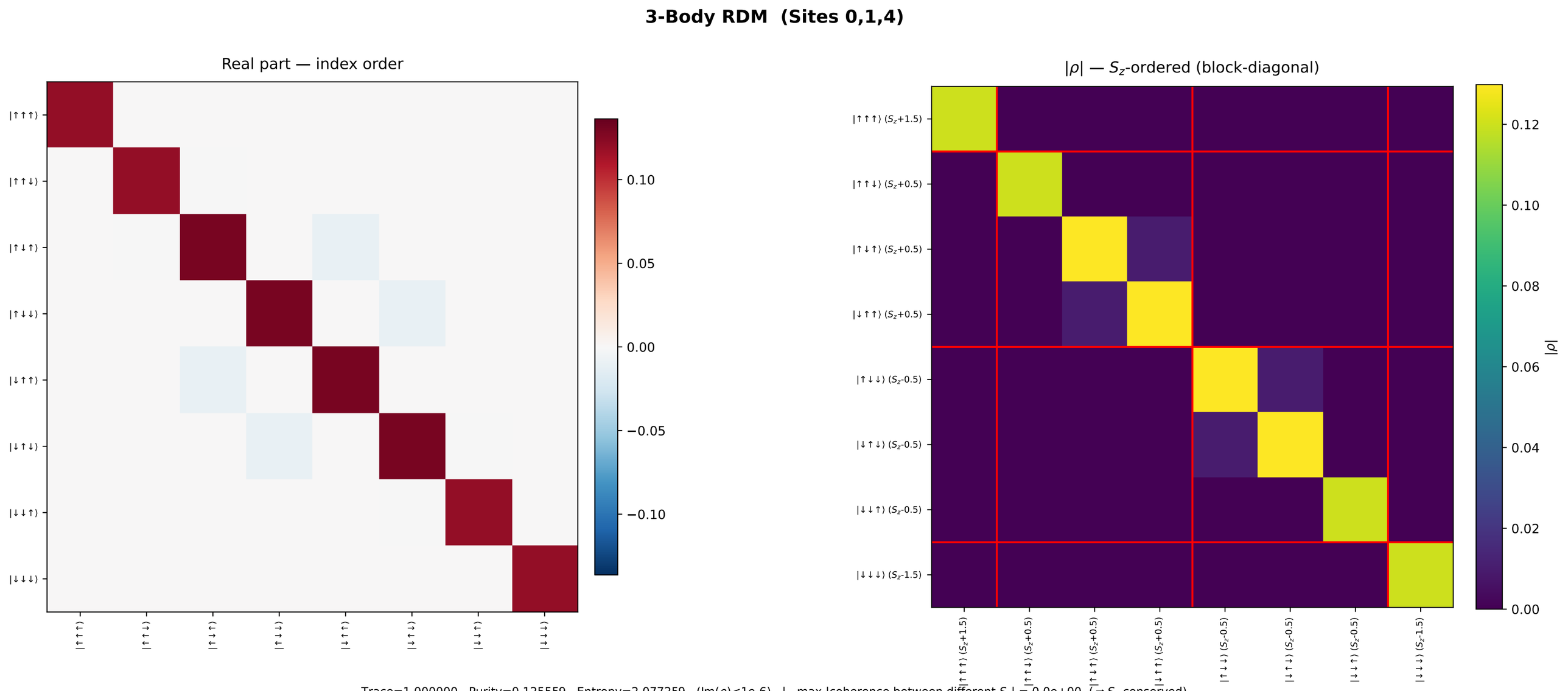
Date: 2026-07-12 22:32

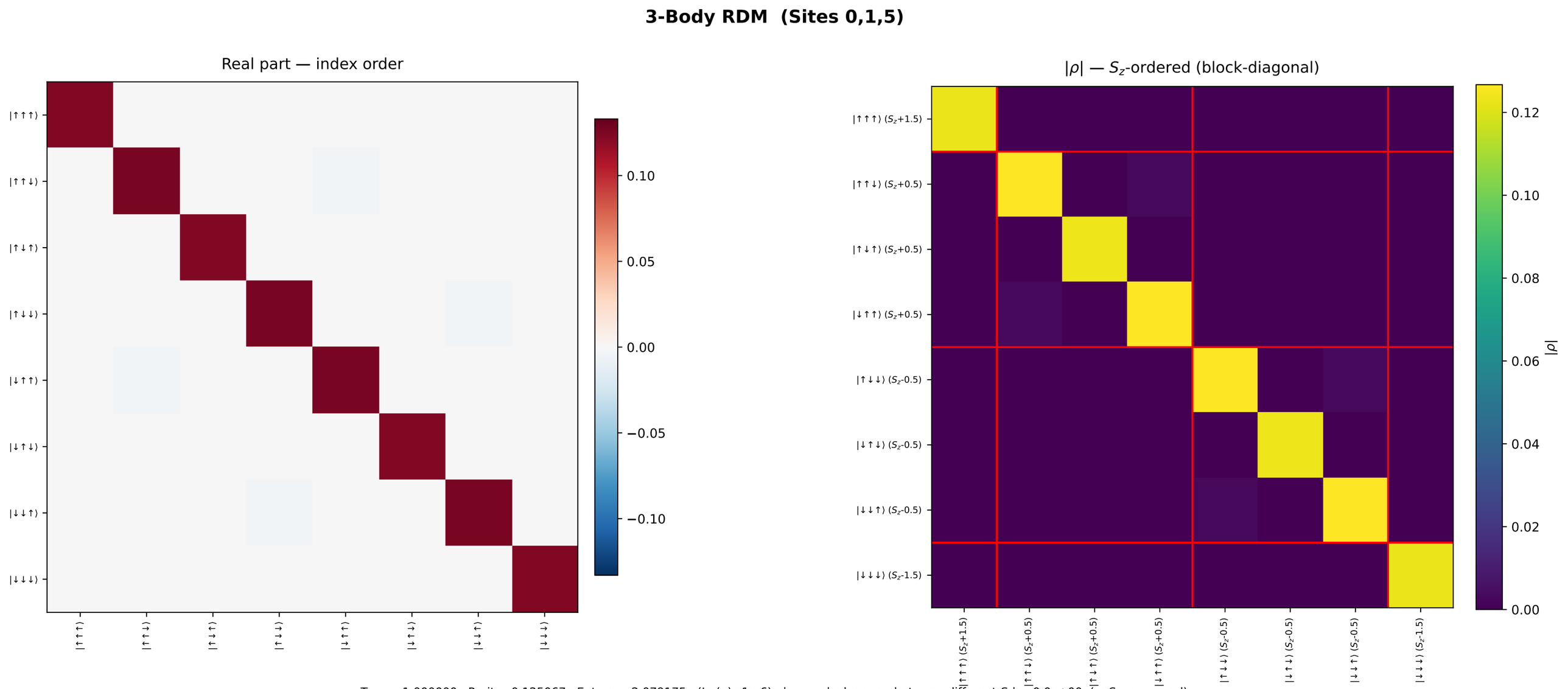
Summary — 3-Body RDMs (Benzene, CAS(6,6))

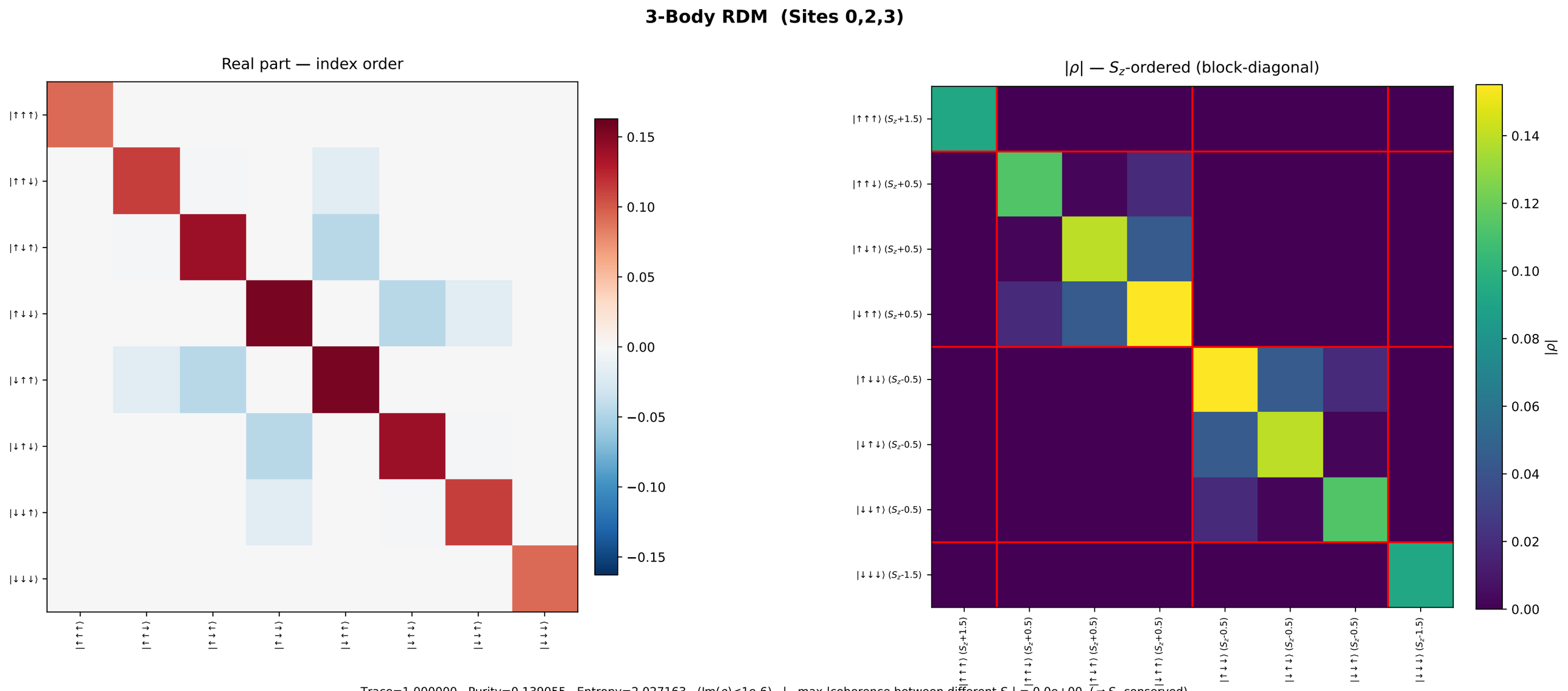
Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.125000	2.079441	—	—	—	—	STRONGLY MIXED
0,1,3	0.170255	1.922402	—	—	—	—	STRONGLY MIXED
0,1,4	0.125559	2.077259	—	—	—	—	STRONGLY MIXED
0,1,5	0.125067	2.079175	—	—	—	—	STRONGLY MIXED
0,2,3	0.139055	2.027163	—	—	—	—	STRONGLY MIXED
0,2,4	0.148904	1.994851	—	—	—	—	STRONGLY MIXED
0,2,5	0.125065	2.079184	—	—	—	—	STRONGLY MIXED
0,3,4	0.125949	2.075505	—	—	—	—	STRONGLY MIXED
0,3,5	0.150430	1.989597	—	—	—	—	STRONGLY MIXED
0,4,5	0.141307	2.019270	—	—	—	—	STRONGLY MIXED
1,2,3	0.151067	1.978814	—	—	—	—	STRONGLY MIXED
1,2,4	0.486887	0.778046	—	—	—	—	MIXED
1,2,5	0.164604	1.941800	—	—	—	—	STRONGLY MIXED
1,3,4	0.499664	0.696539	—	—	—	—	MIXED
1,3,5	0.145172	2.007503	—	—	—	—	STRONGLY MIXED
1,4,5	0.156026	1.961300	—	—	—	—	STRONGLY MIXED
2,3,4	0.499782	0.695444	—	—	—	—	MIXED
2,3,5	0.125967	2.075664	—	—	—	—	STRONGLY MIXED
2,4,5	0.144255	2.010634	—	—	—	—	STRONGLY MIXED
3,4,5	0.125368	2.077991	—	—	—	—	STRONGLY MIXED

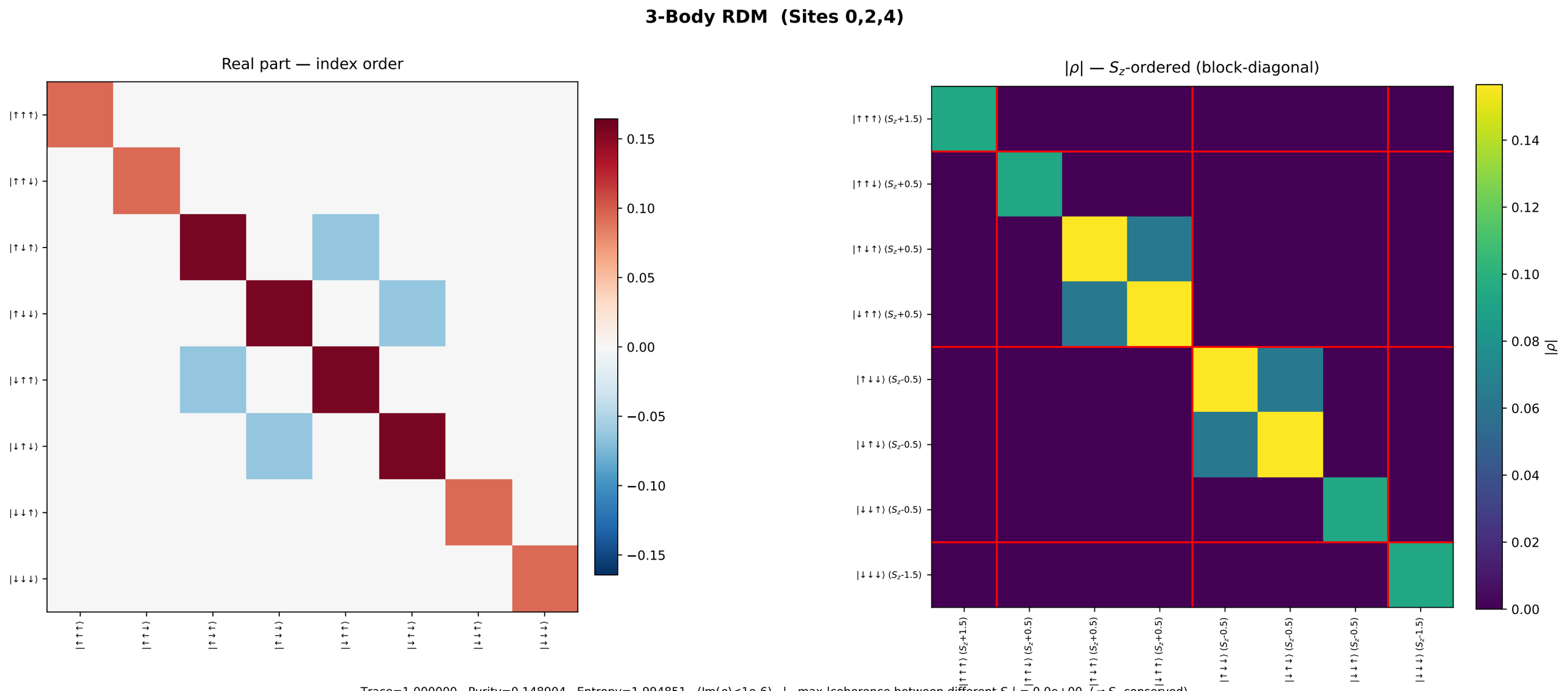

3-Body RDM (Sites 0,1,2)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

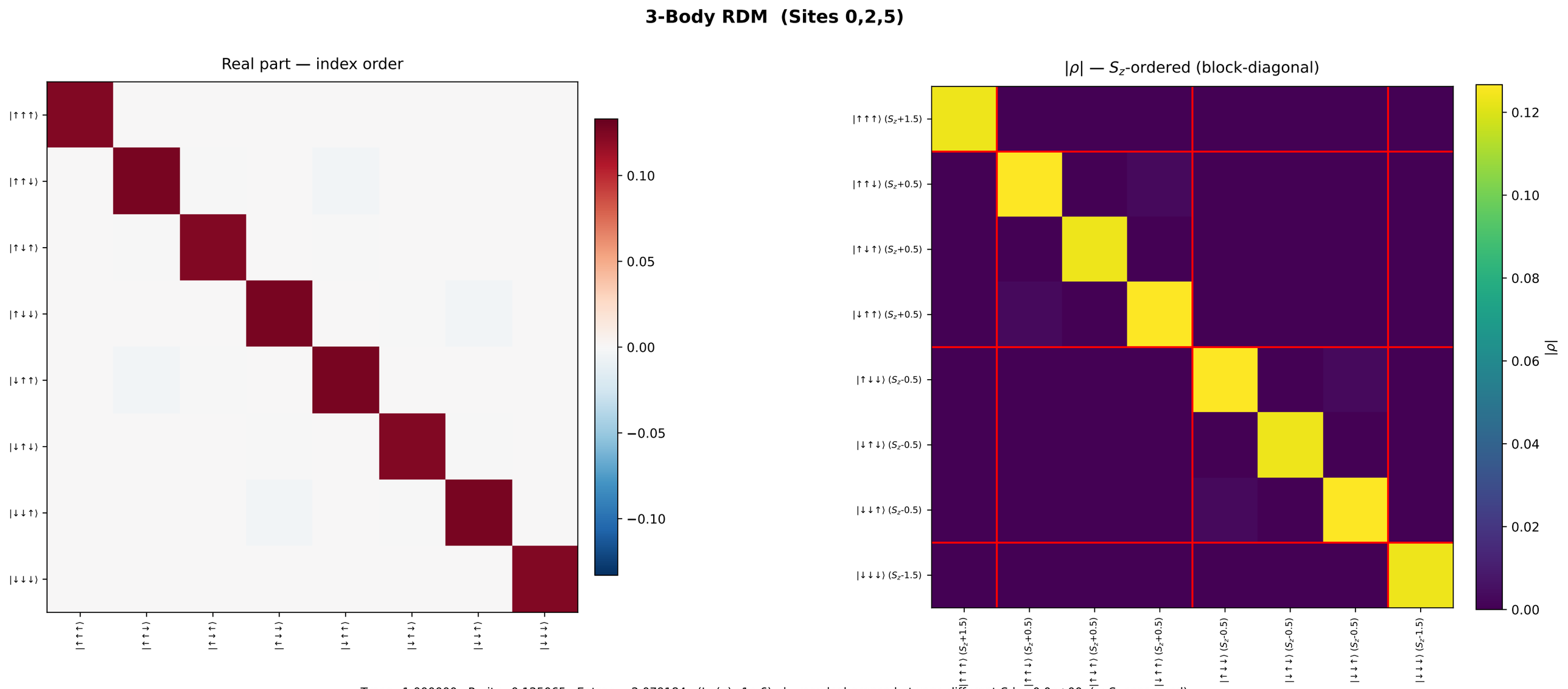

3-Body RDM (Sites 0,1,3)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.170255 Entropy=1.922402 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

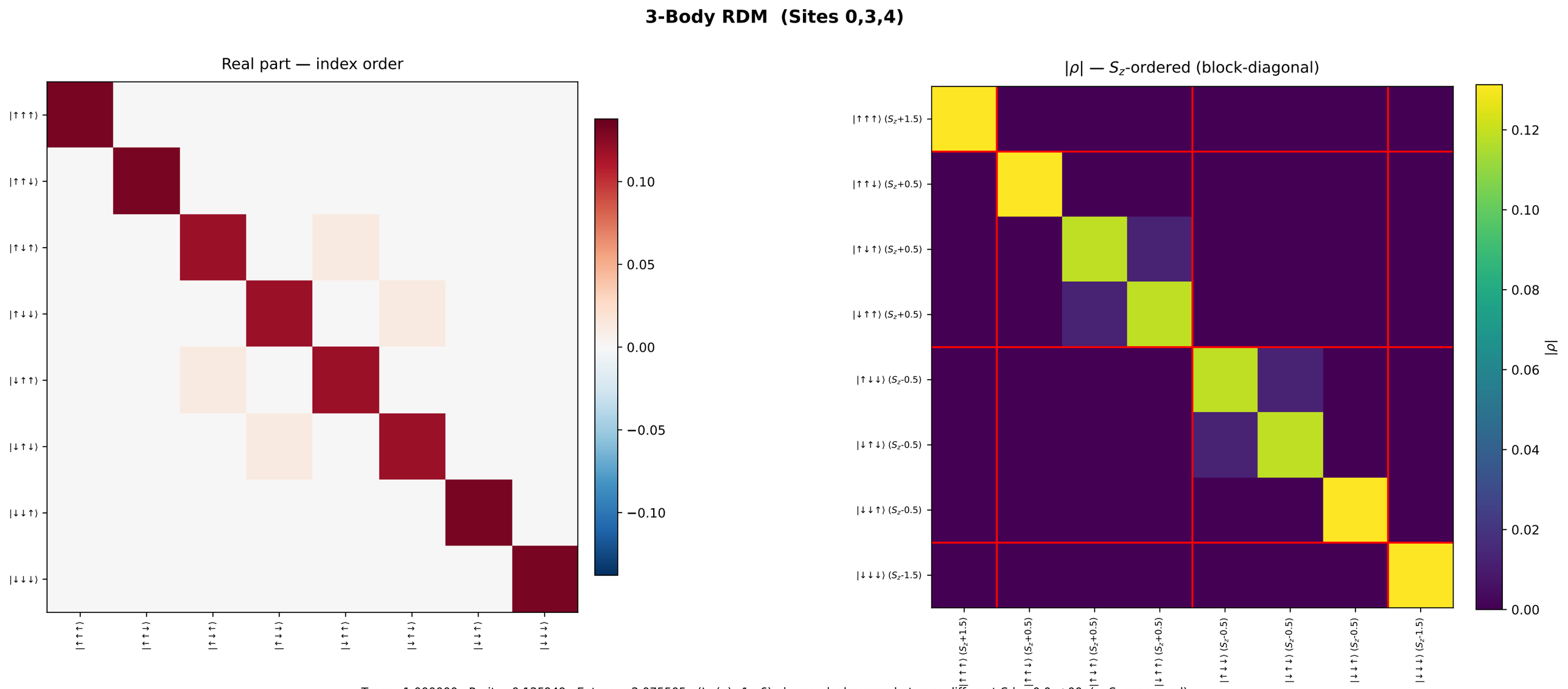

3-Body RDM (Sites 0,1,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125559 Entropy=2.077259 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

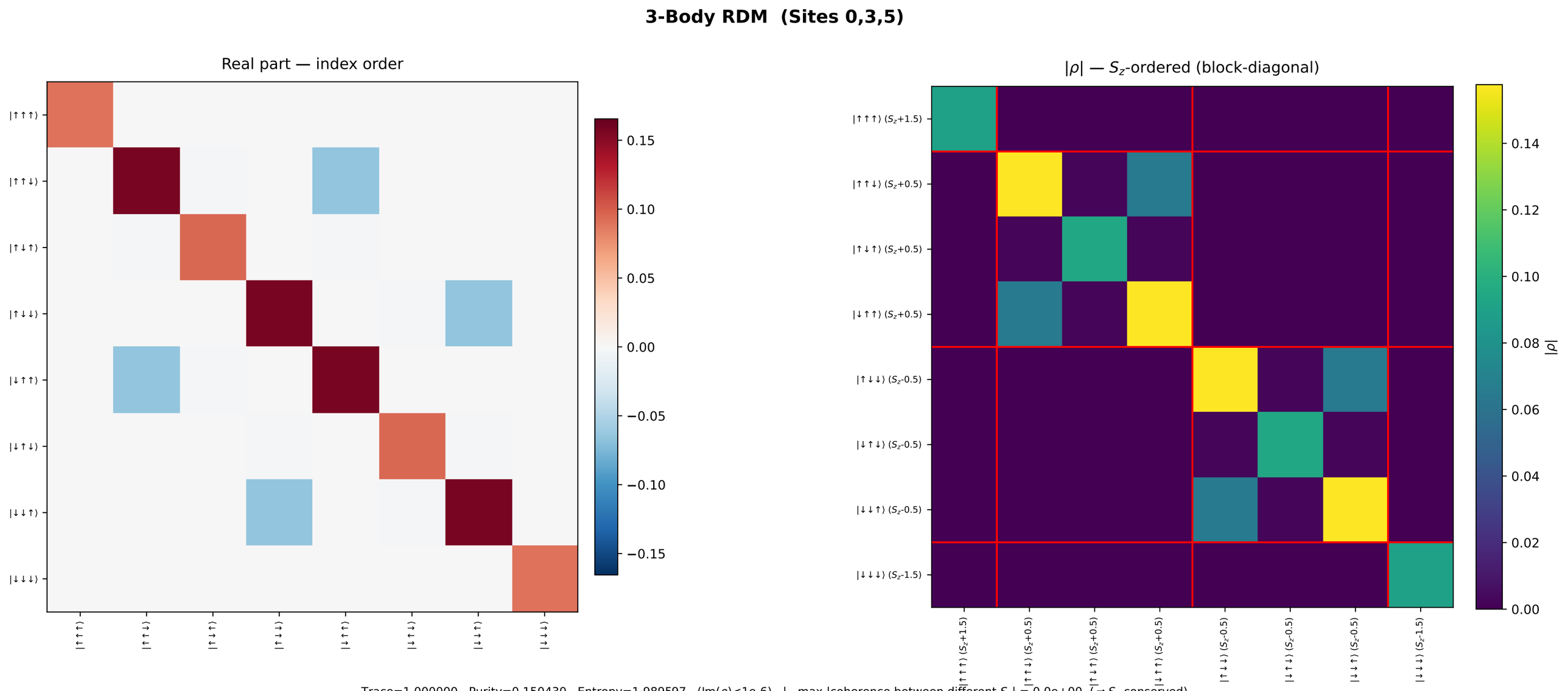

3-Body RDM (Sites 0,1,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125067 Entropy=2.079175 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 0,2,3)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.139055 Entropy=2.027163 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

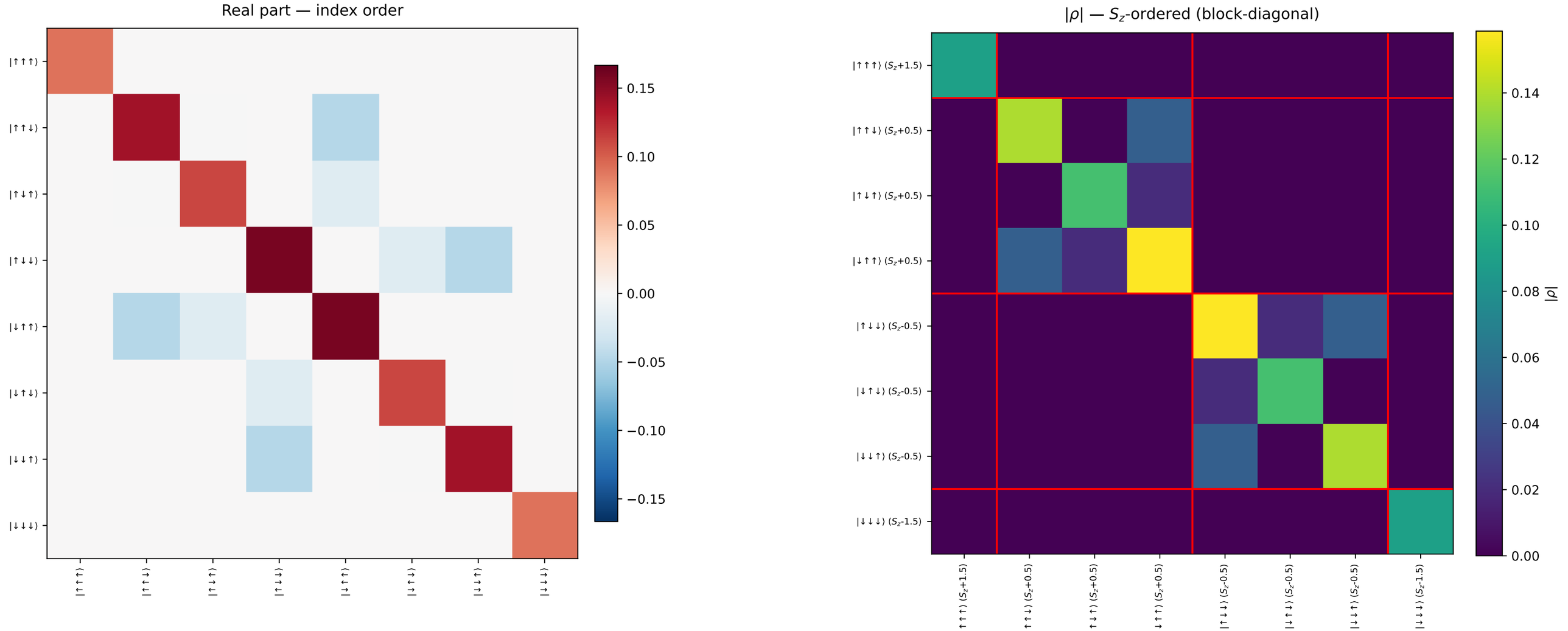

3-Body RDM (Sites 0,2,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.148904 Entropy=1.994851 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 0,2,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125065 Entropy=2.079184 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

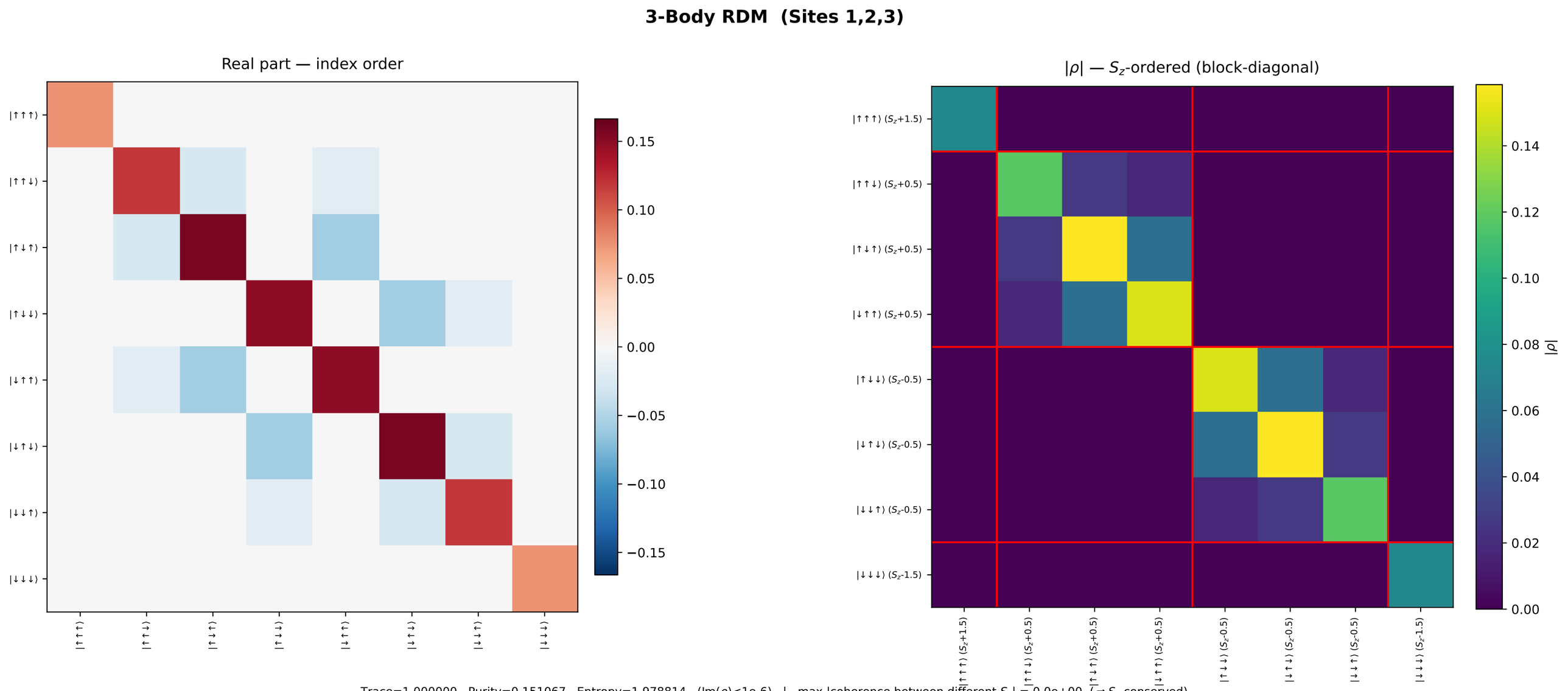

3-Body RDM (Sites 0,3,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125949 Entropy=2.075505 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 0,3,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.150430 Entropy=1.989597 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

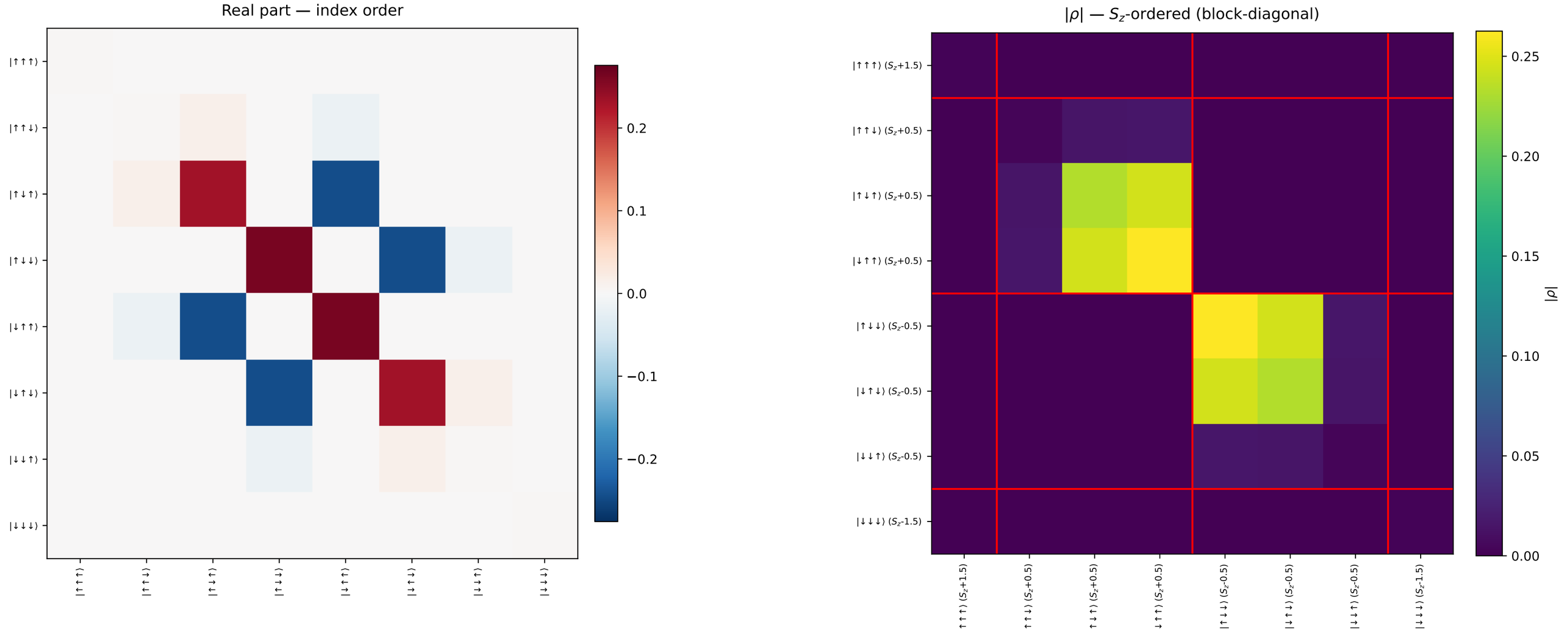
3-Body RDM (Sites 0,4,5)



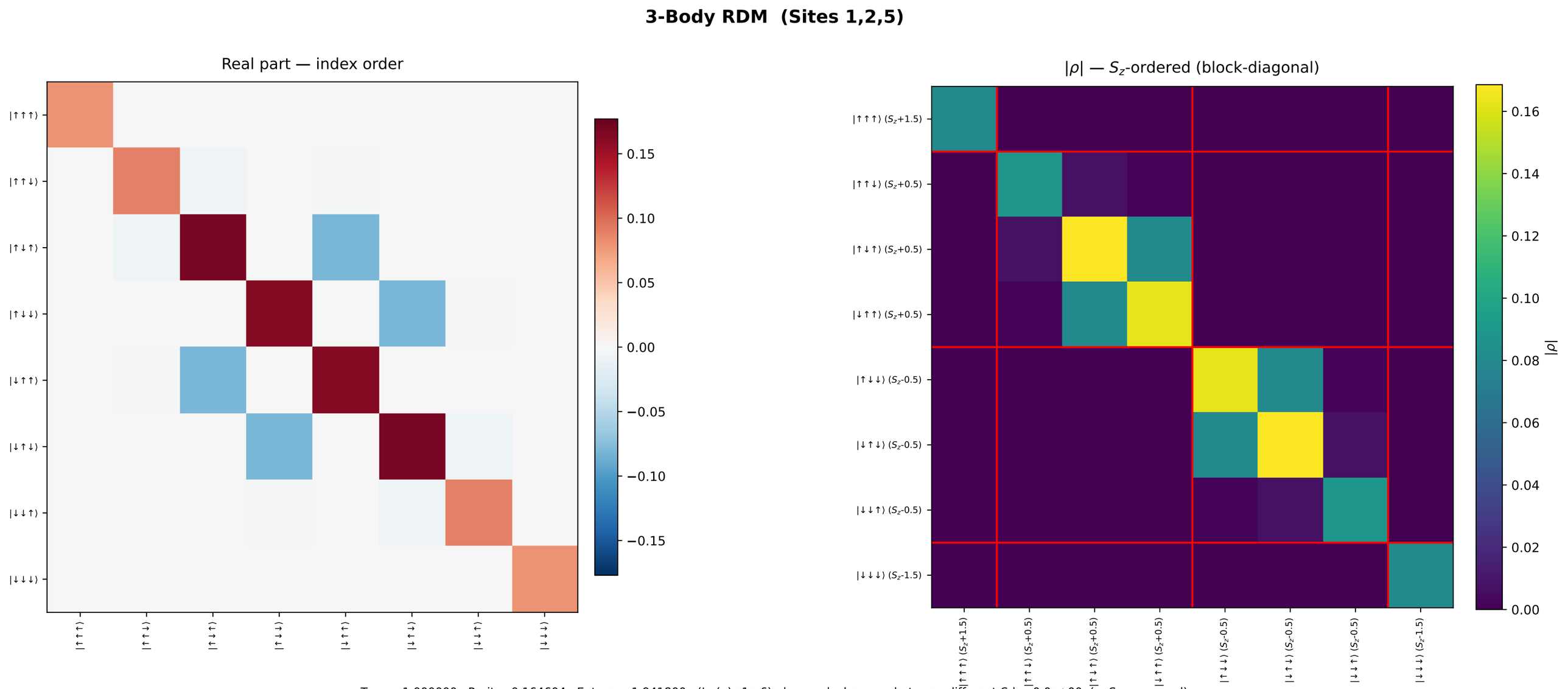
Trace=1.000000 Purity=0.141307 Entropy=2.019270 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

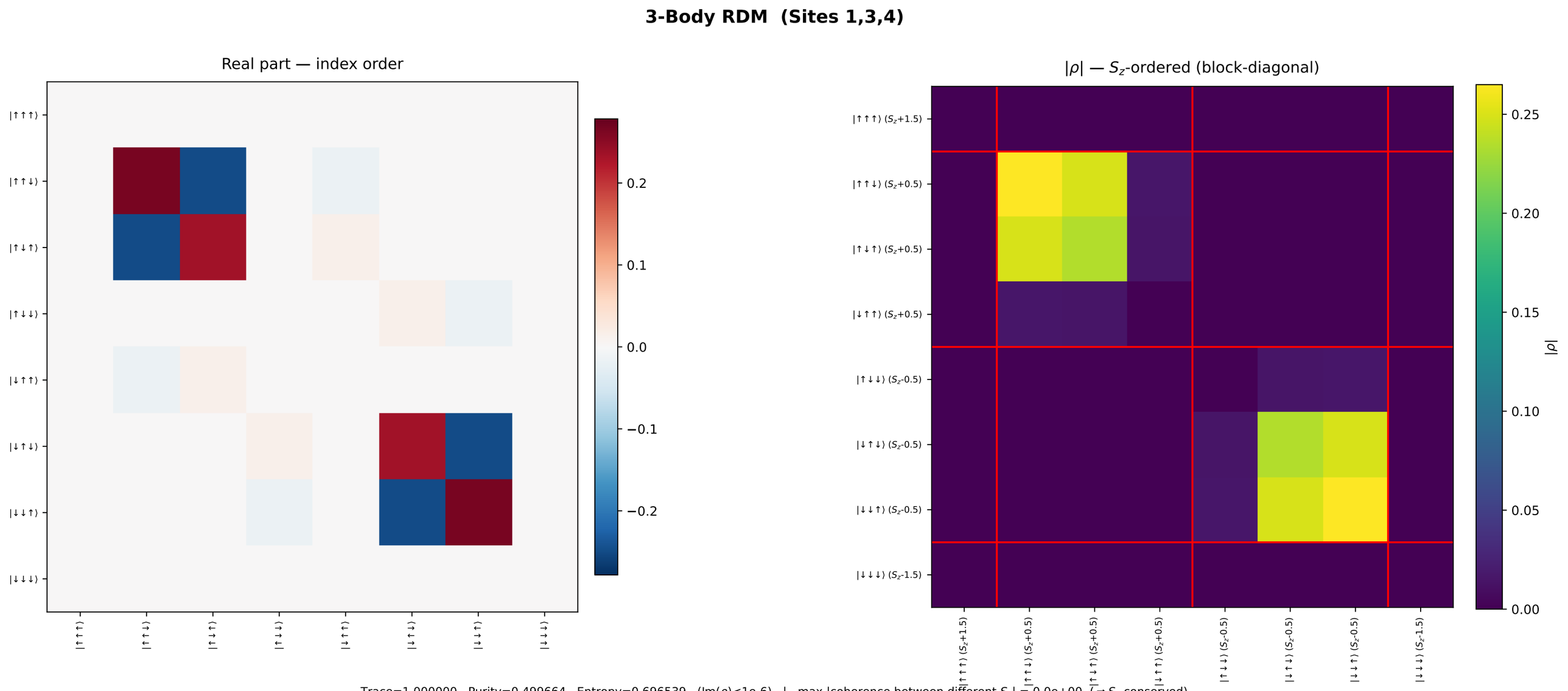

3-Body RDM (Sites 1,2,3)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.151067 Entropy=1.978814 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

3-Body RDM (Sites 1,2,4)

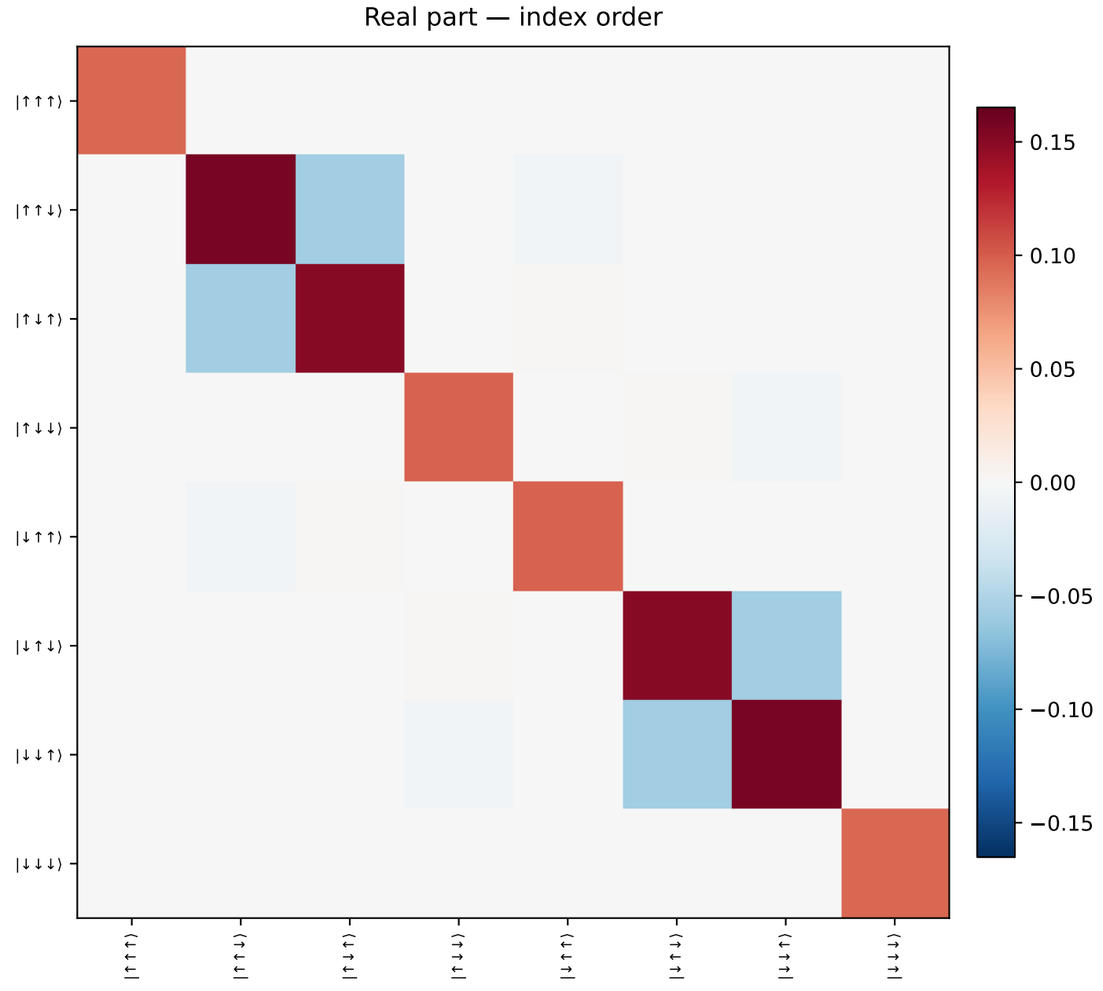


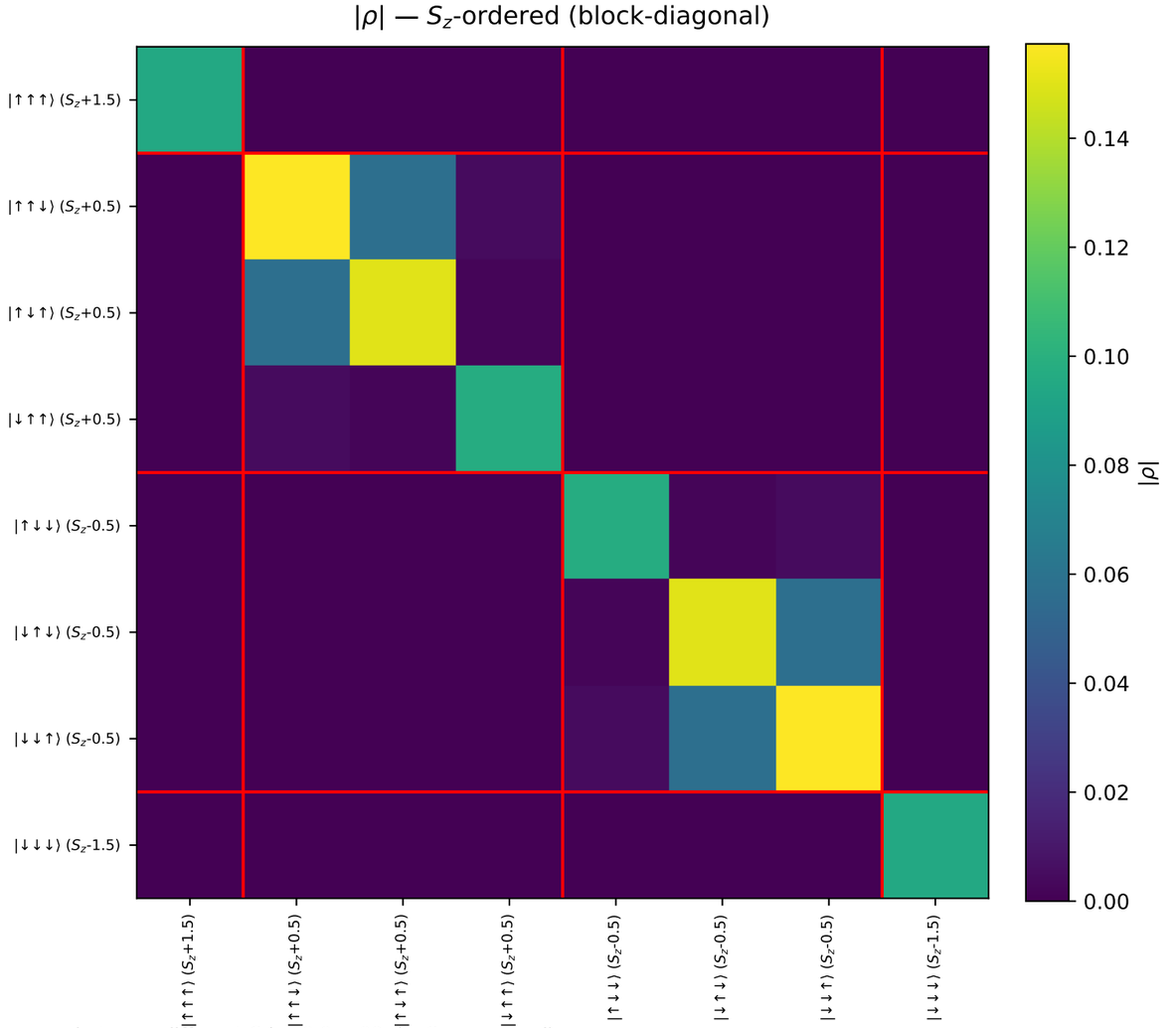
Trace=1.000000 Purity=0.486887 Entropy=0.778046 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)


3-Body RDM (Sites 1,2,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.164604 Entropy=1.941800 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 1,3,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.499664 Entropy=0.696539 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

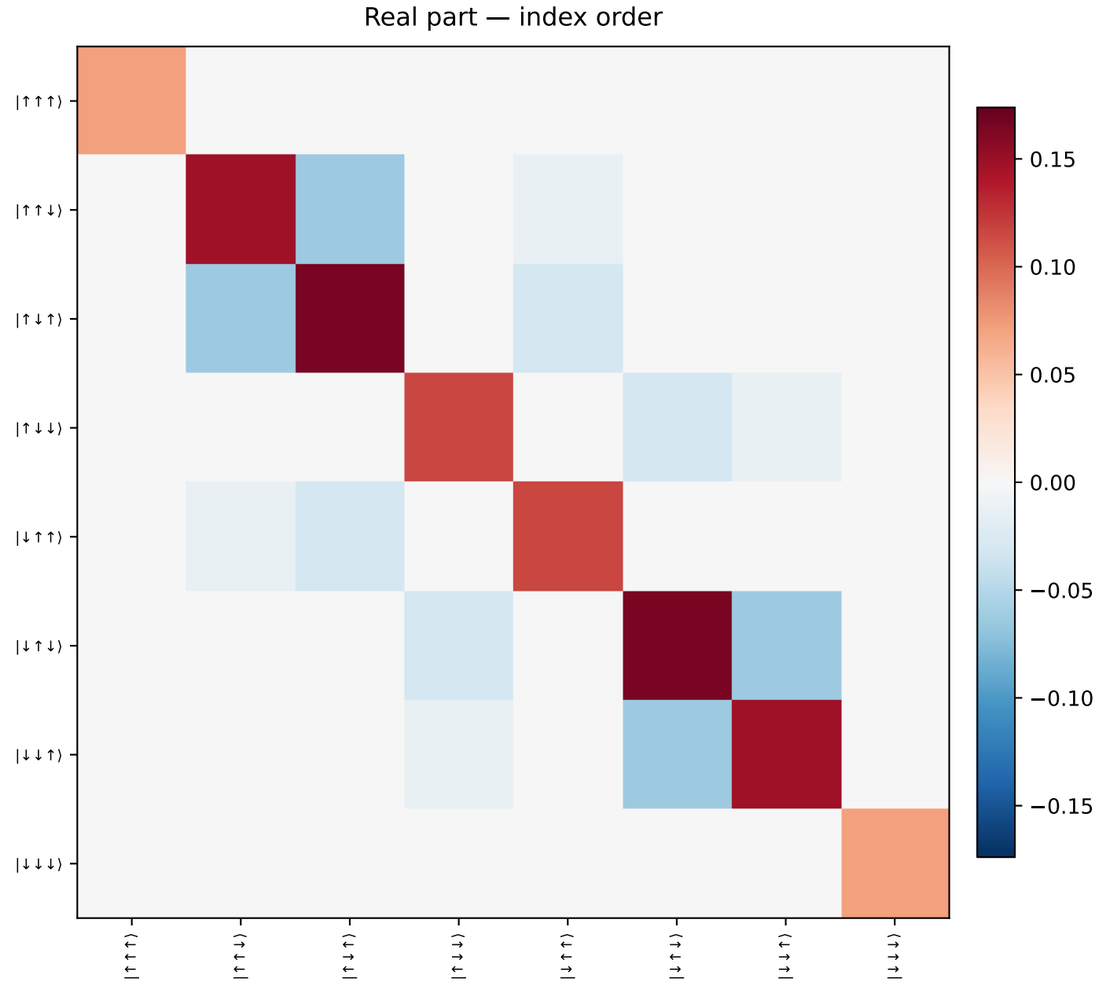
3-Body RDM (Sites 1,3,5)

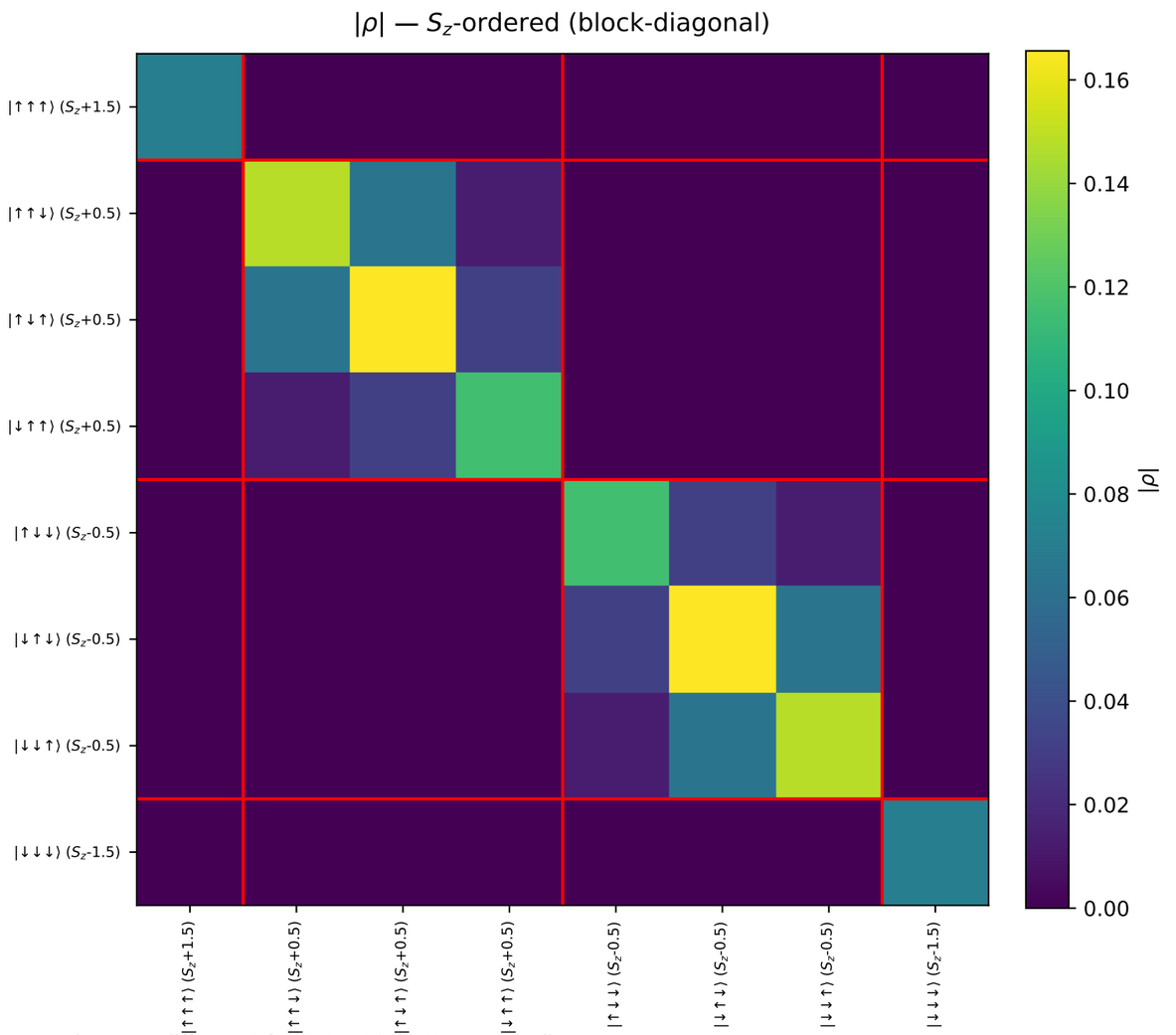




Trace=1.000000 Purity=0.145172 Entropy=2.007503 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

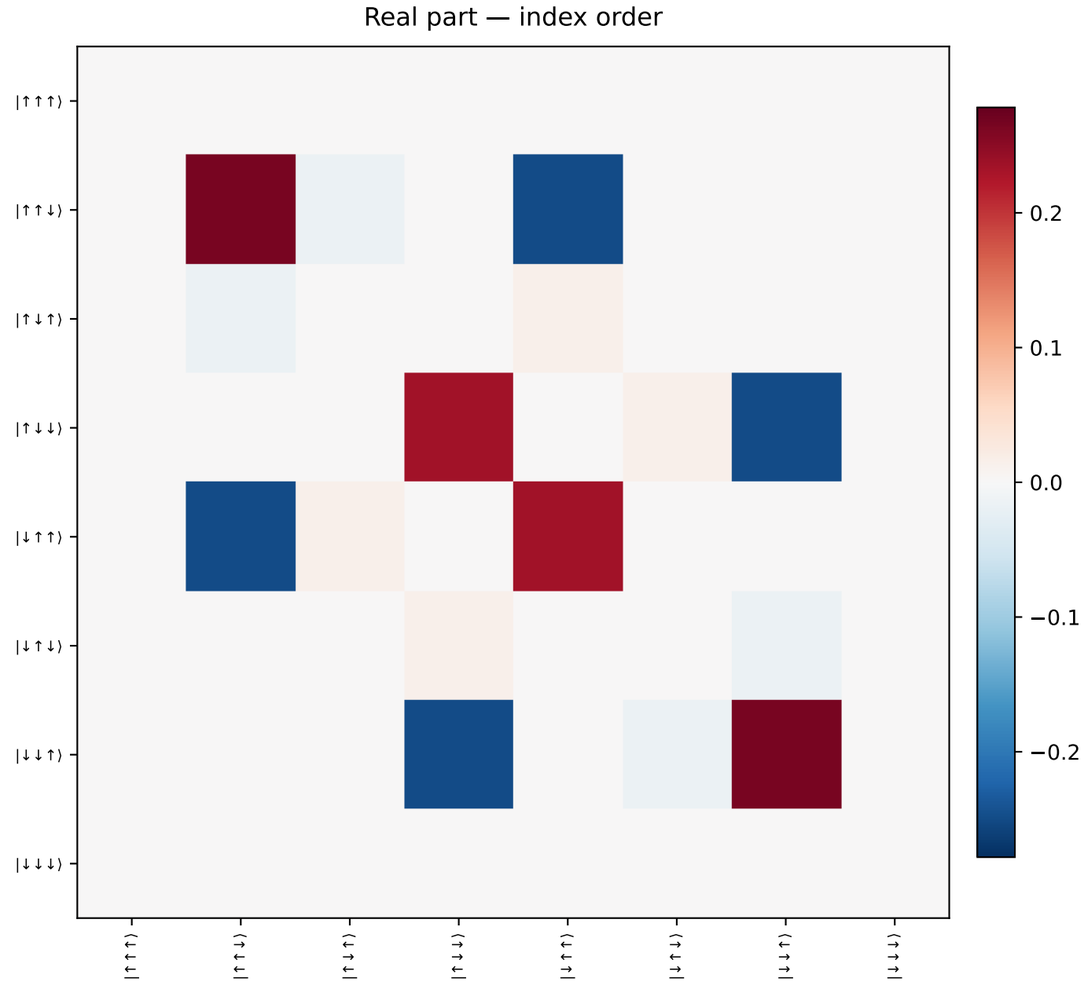
3-Body RDM (Sites 1,4,5)

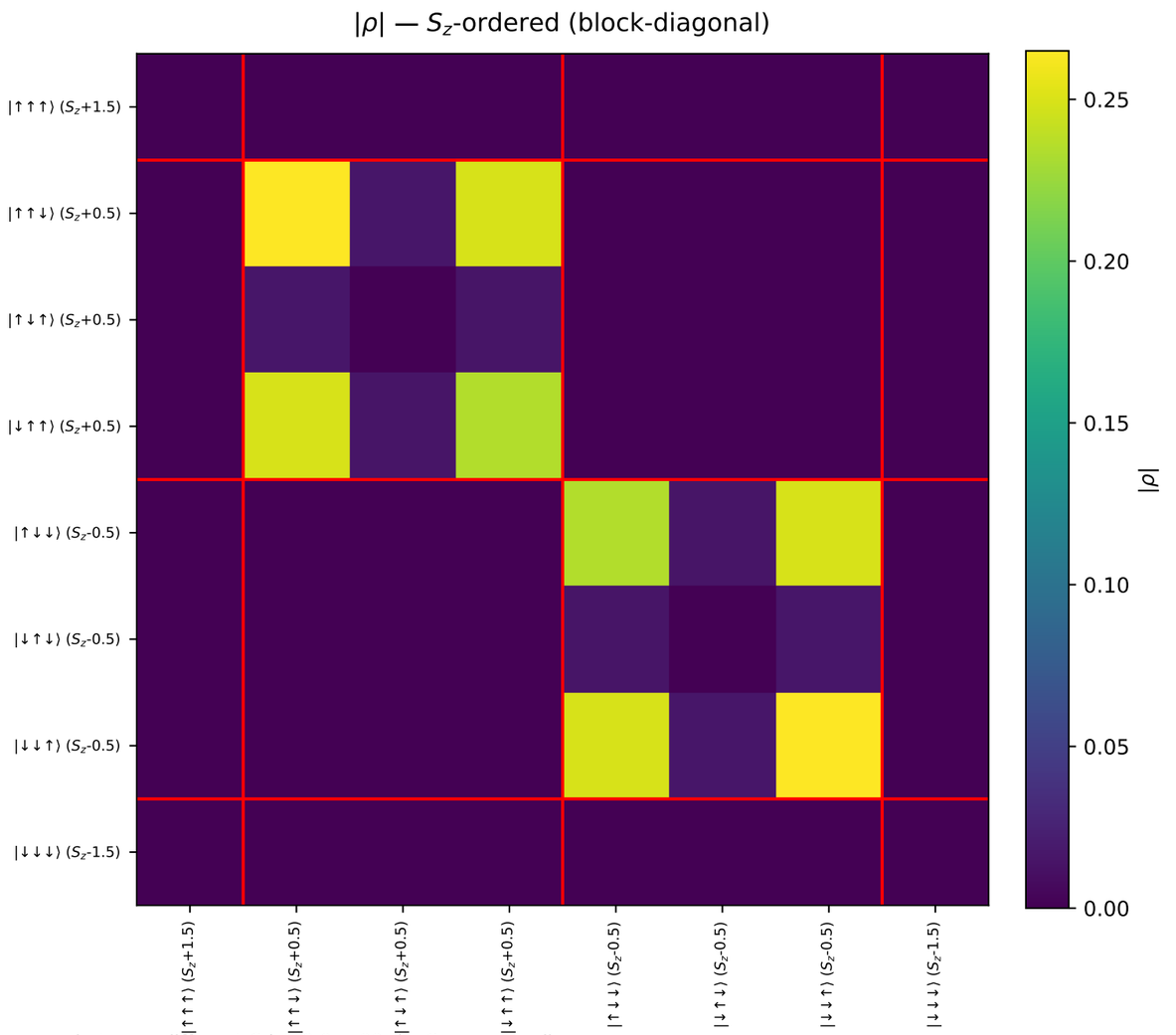




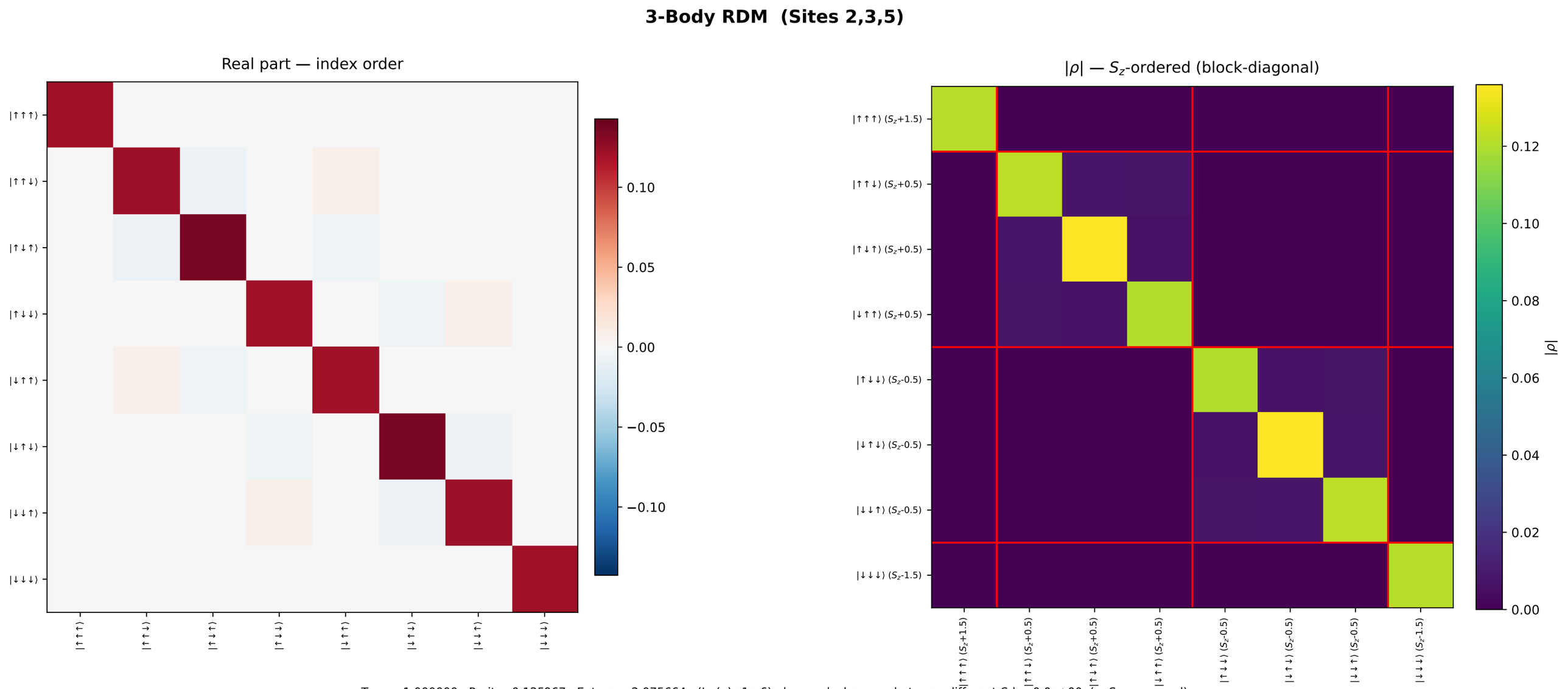
Trace=1.000000 Purity=0.156026 Entropy=1.961300 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

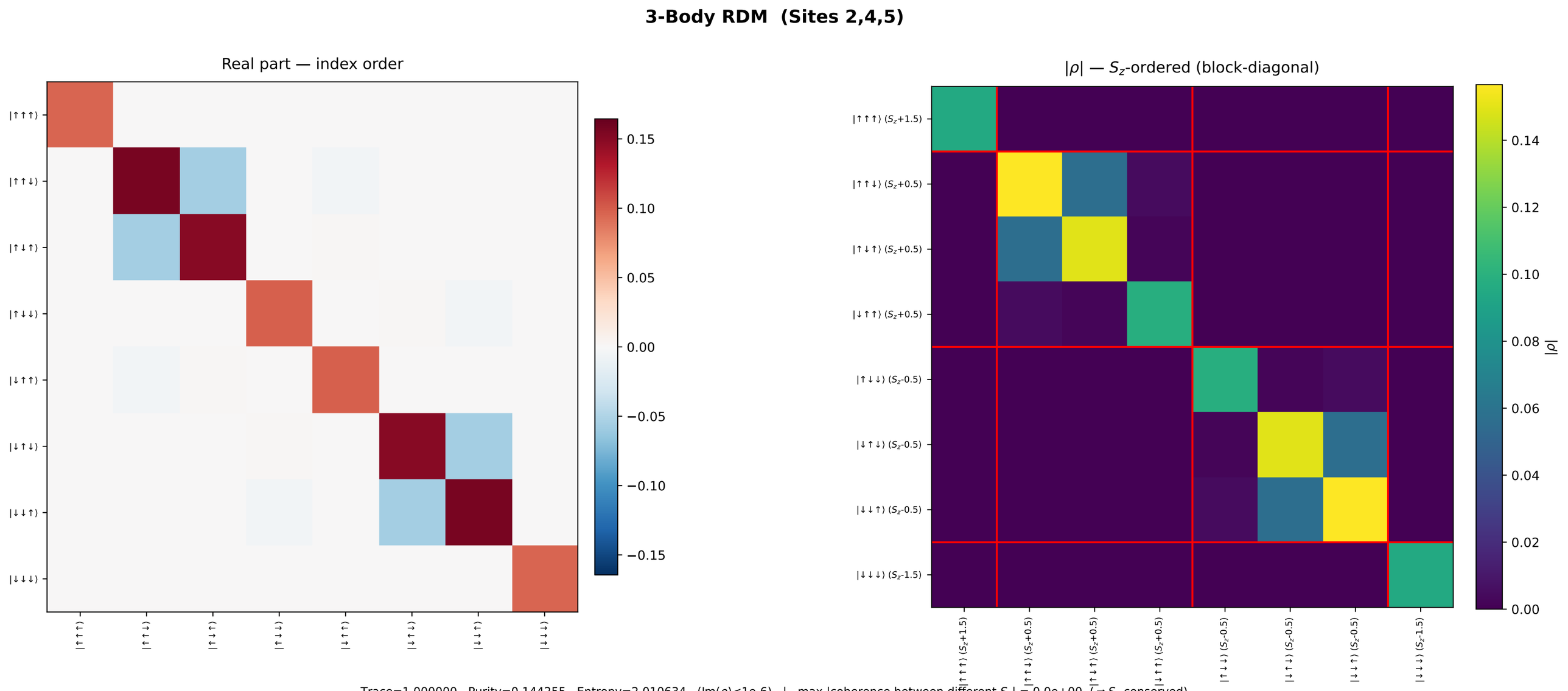
3-Body RDM (Sites 2,3,4)



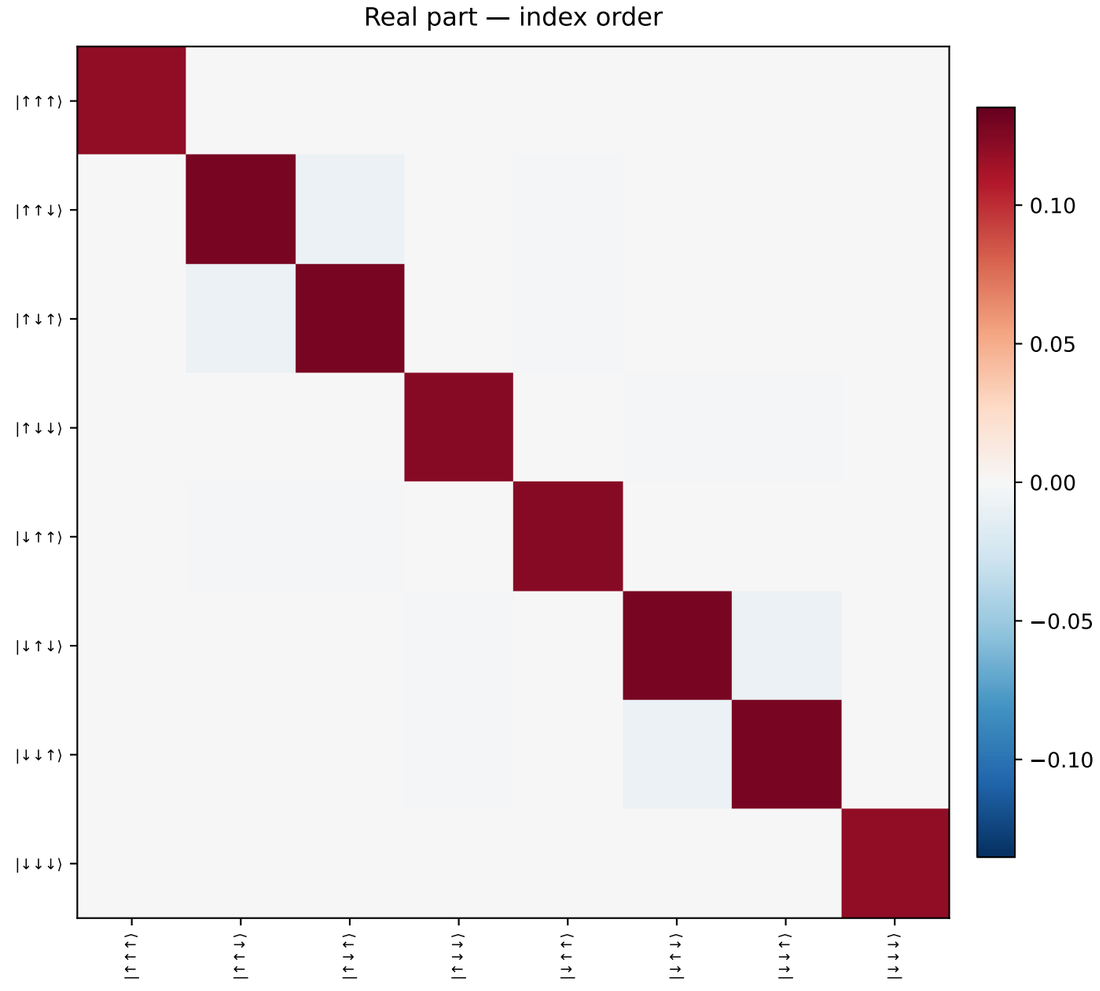


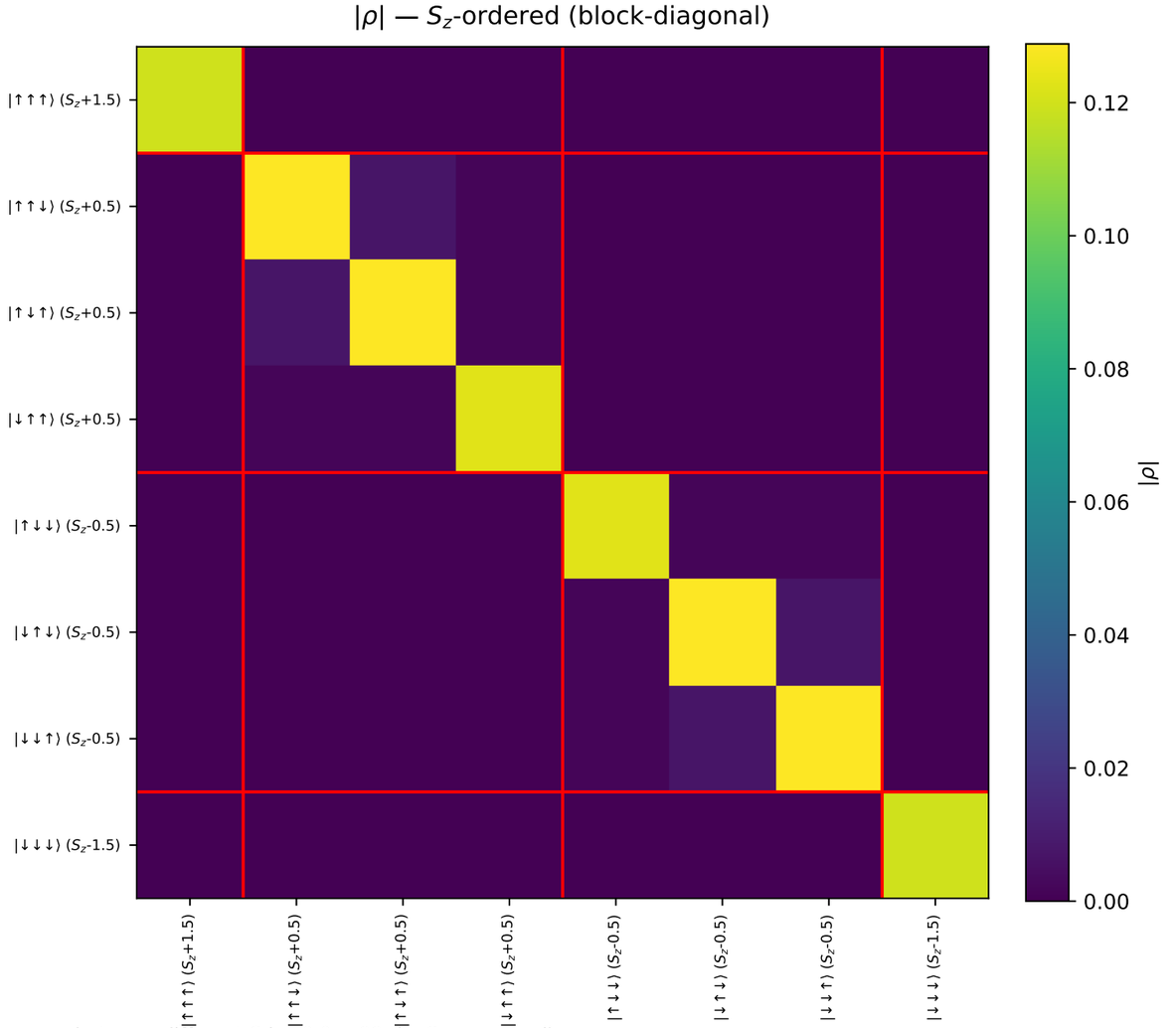
Trace=1.000000 Purity=0.499782 Entropy=0.695444 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)


3-Body RDM (Sites 2,3,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125967 Entropy=2.075664 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 2,4,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.144255 Entropy=2.010634 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

3-Body RDM (Sites 3,4,5)





Trace=1.000000 Purity=0.125368 Entropy=2.077991 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)